AN ALGEBRAIC SPACE ETALE OVER THE SPECTRUM OF A FIELD IS A SCHEME

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied verbatim from the pinned Stacks Project chapter file spaces-properties.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. Span recomputation by label pairing (LANE_BRIEF 10.2): in this chapter the author's proof environment follows the statement environment as a standalone block; the spec covers the complete environment pair (statement span, proof span) as recorded in the pool candidate.

Editorial note. Occupation scan: combined claim_key + statement_anchor + generic-line scan over all registration files (active and withdrawn) found 13 S10-geom published cards for spaces-properties.tex (prop-occupied.json, 2026-10-14); those 13 labels are excluded from this batch by design. For the 77 free labels: 0 occupied. The registry mentions field (7557 files) and focused-seeds (645) both point to the same S10-geom cards (dual-source check), no other occupation.

Source reference. Original definition 03FR: sources/stacks-chow/tags.txt, source lines 4502–4502. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Source reference. Original reference 02GL: sources/stacks-chow/tags.txt, source lines 3232–3232. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Source reference. Original reference 03DP: sources/stacks-chow/tags.txt, source lines 4428–4428. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Source reference. Original reference 03BM: sources/stacks-chow/tags.txt, source lines 4353–4353. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Source reference. Original reference 02WU: sources/stacks-chow/tags.txt, source lines 3817–3817. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Source reference. Original lemma 03JH: sources/stacks-chow/tags.txt, source lines 4639–4639. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Lemma 0.1. *Let S be a scheme. Let $X \to \mathrm{Spec}(k)$ be étale morphism over S, where k is a field. Then X is a scheme.*

Proof. Let U be an affine scheme, and let $U \to X$ be an étale morphism. By Definition 03FR we see that $U \to \mathrm{Spec}(k)$ is an étale morphism. Hence $U = \coprod_{i=1,\ldots,n} \mathrm{Spec}(k_i)$ is a finite disjoint union of spectra of finite separable extensions k_i of k, see Morphisms, Lemma 02GL. The $R = U \times_X U \to U \times_{\mathrm{Spec}(k)} U$ is a

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monomorphism and $U \times_{\mathrm{Spec}(k)} U$ is also a finite disjoint union of spectra of finite separable extensions of k. Hence by Schemes, Lemma 03DP we see that R is similarly a finite disjoint union of spectra of finite separable extensions of k. This U and R are affine and both projections $R \to U$ are finite locally free. Hence U/R is a scheme by Groupoids, Proposition 03BM. By Spaces, Lemma 02WU it is also an open subspace of X. By Lemma 03JH we conclude that X is a scheme. $\square$